EMERGENCY_STOP
0: STOP
1: GO

CORD

ON/OFF
0: OFF
1: ON

TEAM

PWM

EYE
Other MCU

LIDAR

BATTERY

OTOS

J103
Conn_01x02

J104
Conn_01x02

J105
Conn_01x02

J106
Conn_01x02

Conn_01x03
J107

+5V_SW

J108
BM04B-SRSS-TB

+3.3V

I2C_EYE_SCL
I2C_EYE_SDA

+5V

JP102

JP101

J109
Conn_01x04

PAMI_nON
PAMI_nSTOP

Power
PAMI_nON
PAMI_nSTOP
VBAT_MEAS
IPAMI_MEAS

File: 0500_Power.kicad_sch

VBAT_IN

J101
Conn_01x02

+3.3V

I2C_OTOS_SCL
I2C_OTOS_SDA

J116
BM04B-SRSS-TB

J102
USB_C_Receptacle_USB2.0_14P

VBUS

CC1
CC2
D-
D-
D+
D+

SHIELD
GND
SH
A1

5.1kΩ
R101
5.1kΩ
R102

IO
DSI_STOP
DSI_CORD_IN
DSI_ON
DSI_TEAM_IN
PWM_S1_OUT
PAMI_STOP
PAMI_nSTOP
DSI_CORD
PAMI_nON
DSI_TEAM
PWM_S1
PWM_LED_RGB

File: 0400_IOEXT.kicad_sch

+3.3V +3.3V
R113 3.3kΩ
R115 3.3kΩ

MCU
DSI_STOP
DSI_CORD
DSI_TEAM
PWM_S1
PWM_LED
I2C_EYE_SCL
I2C_EYE_SDA
LIDAR_UART_TX
VBAT_MEAS
IPAMI_MEAS
I2C_OTOS_SCL
I2C_OTOS_SDA
USB_D-
USB_D+
I2C_CAM_SDA
I2C_CAM_SCL
M1_STEP
M1_DIR
MOT_UART_TX
MOT_UART_RX
M2_STEP
M2_DIR
M3_STEP
M3_DIR
M_nEN

File: 0100_MCU.kicad_sch

+3.3V +3.3V
R109 3.3kΩ
R110 3.3kΩ

R111 1kΩ
R112 0Ω

+3.3V +3.3V
R105 100kΩ
R107 100kΩ

R103 100kΩ
R104 100kΩ
R106 100kΩ
R108 100kΩ

I2C_CAM
I2C_CAM_SDA
I2C_CAM_SCL
I2C_CAM_SCL0
I2C_CAM_SDA0
I2C_CAM_SCL1
I2C_CAM_SDA1
I2C_CAM_SCL2
I2C_CAM_SDA2

File: 0300_I2C_CAM.kicad_sch

+5V
+5V
+5V

BM04B-SRSS-TB
J110
CAM1

BM04B-SRSS-TB
J111
CAM2

BM04B-SRSS-TB
J112
CAM3

Motor1
STEPPER_STEP
STEPPER_DIR
UART_RTX
UART_AD0
UART_AD1
STEPPER_1B
STEPPER_1A
STEPPER_2A
STEPPER_2B
DRV_nEN

File: 0200_Motors.kicad_sch

J113
Conn_01x04
MOTOR1

Motor2
STEPPER_STEP
STEPPER_DIR
UART_RTX
UART_AD0
UART_AD1
STEPPER_1B
STEPPER_1A
STEPPER_2A
STEPPER_2B
DRV_nEN

File: 0200_Motors.kicad_sch

J114
Conn_01x04
MOTOR2

Motor3
STEPPER_STEP
STEPPER_DIR
UART_RTX
UART_AD0
UART_AD1
STEPPER_1B
STEPPER_1A
STEPPER_2A
STEPPER_2B
DRV_nEN

File: 0200_Motors.kicad_sch

J115
Conn_01x04
MOTOR3

D101
BAV70-C68978
PAMI_STOP
M_nEN

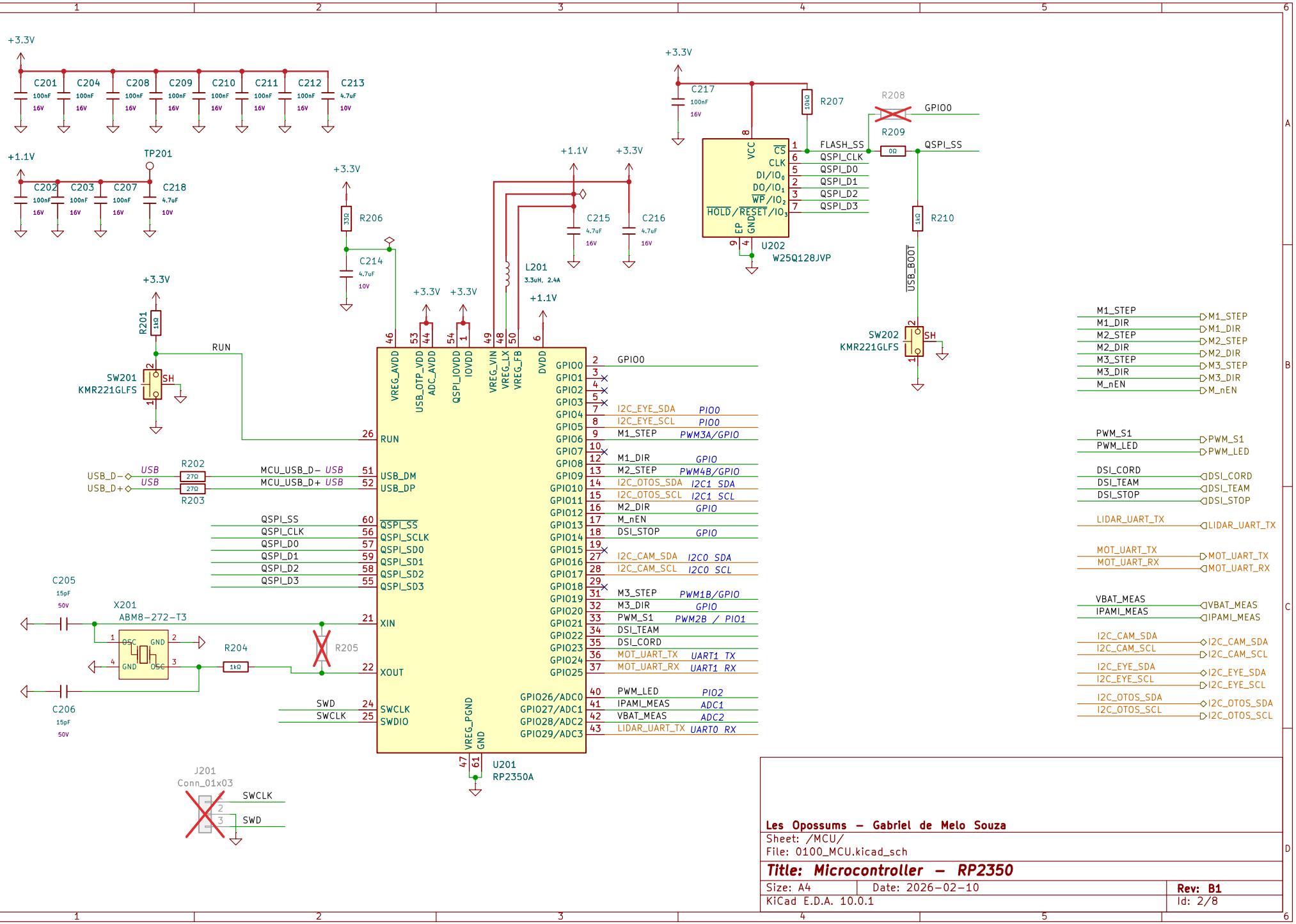
JP103

**Les Opossums – Gabriel de Melo Souza**

Sheet: /
File: PAMI_2027_A1.kicad_sch

***Title: PAMI – Main***

| Size: A3 | Date: 2026-02-10 | **Rev: B1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 1/8 |




Les Opossums – Gabriel de Melo Souza

Sheet: /MCU/

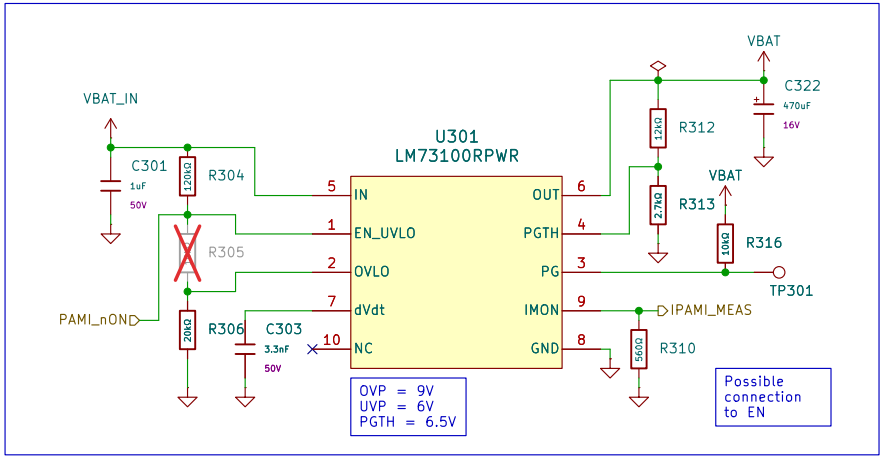
File: 0100_MCU.kicad_sch

**Title: Microcontroller – RP2350**

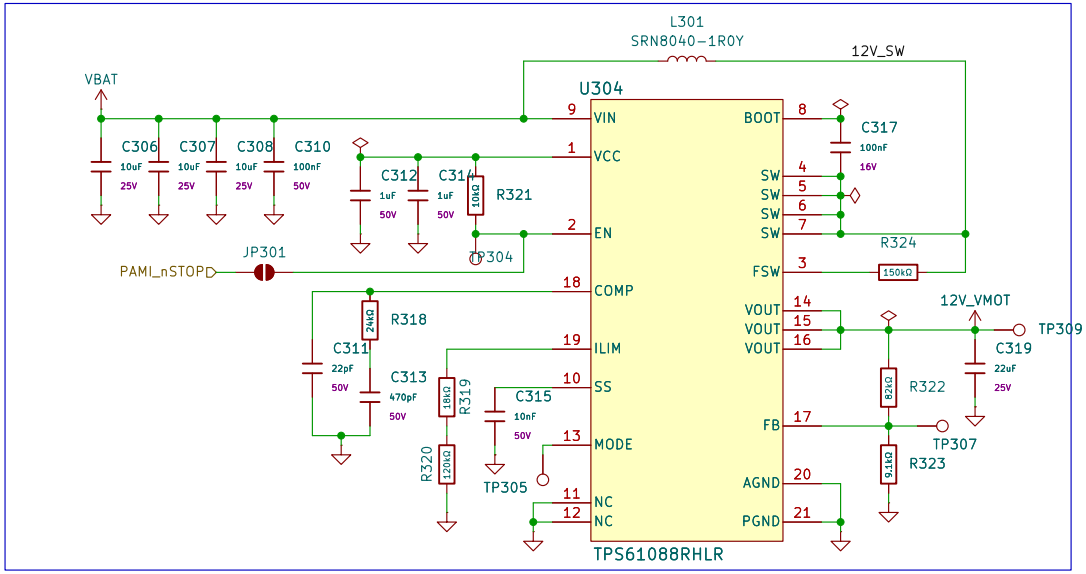
Size: A4 | Date: 2026-02-10 | **Rev: B1**

KiCad E.D.A. 10.0.1 | Id: 2/8

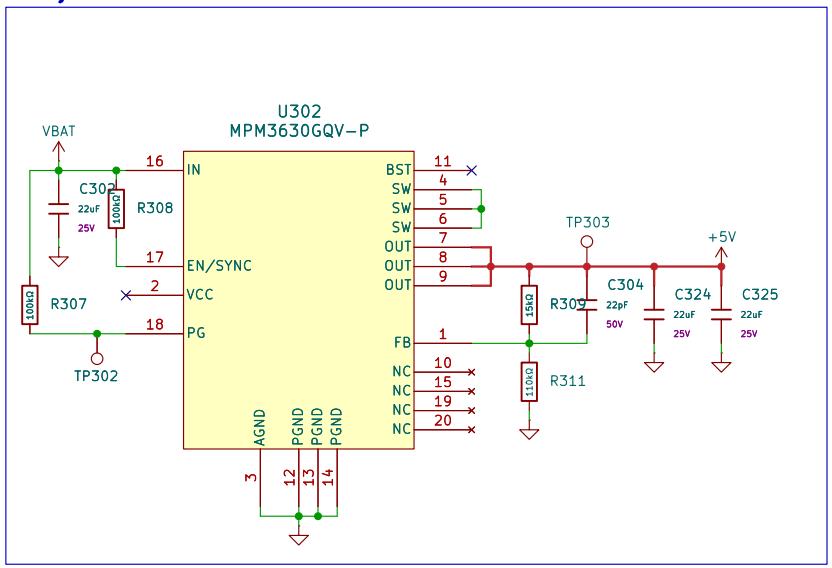
## Load switch – VBAT



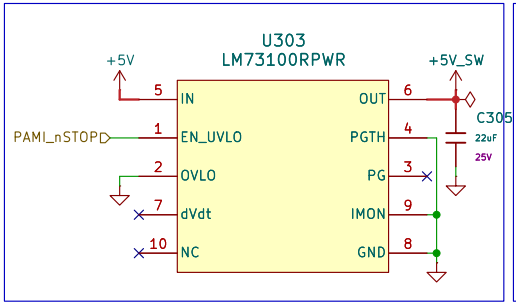

## +12V Stepper Motors generation



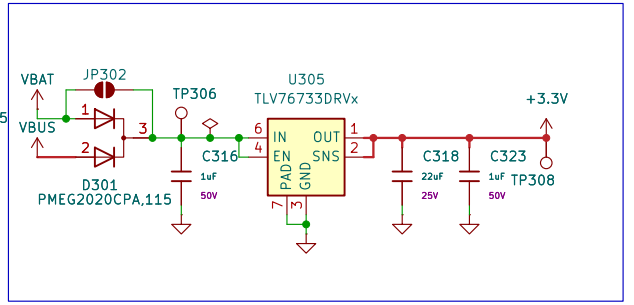

## Always ON 5V LDO



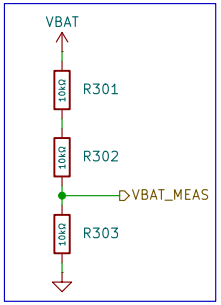

## Switched 5V – +5V_SW




## Always ON 3.3V LDO




## VBAT Measurement





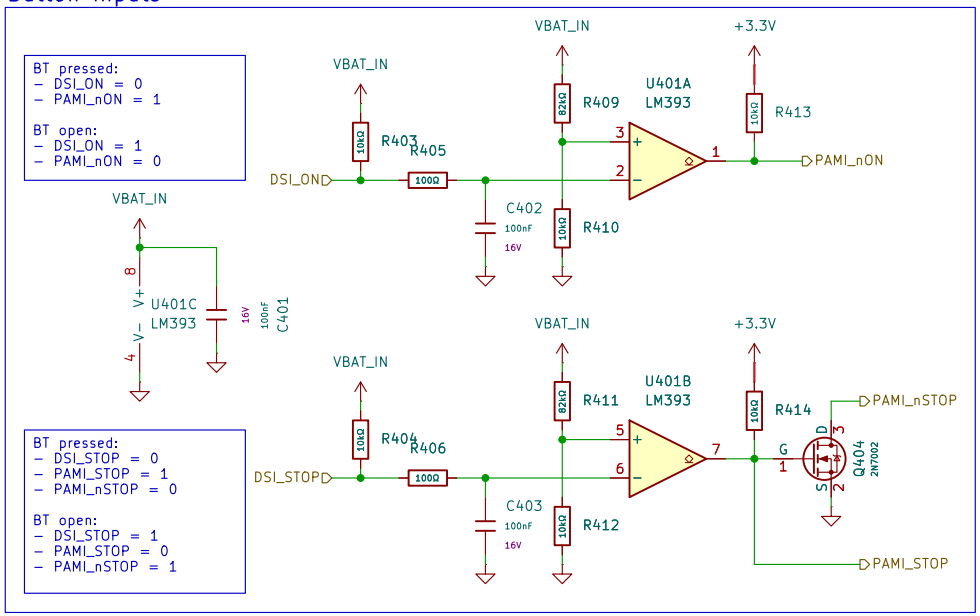
**Les Opossums – Gabriel de Melo Souza**


Sheet: /Power/
File: 0500_Power.kicad_sch

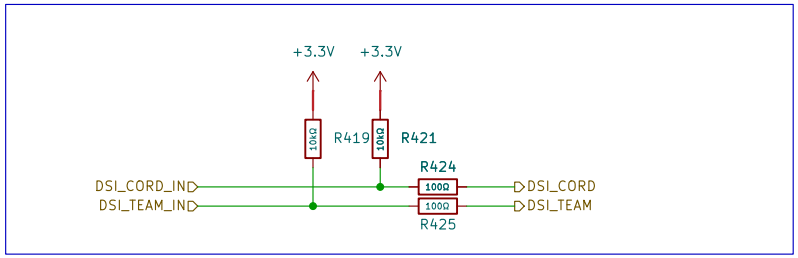
**Title: Power**

| Size: A4 | Date: 2026-02-10 | **Rev: B1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 3/8 |

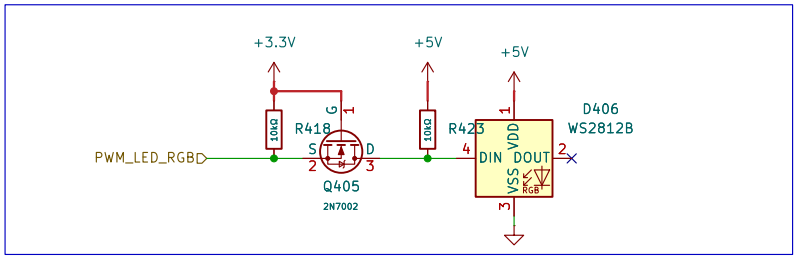
## Button inputs



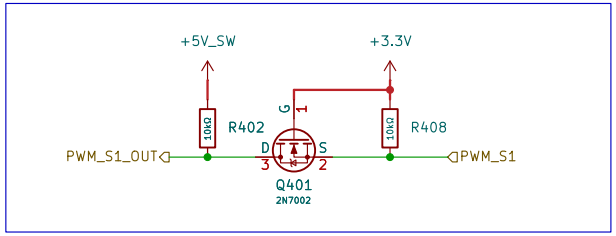

## Discrete inputs



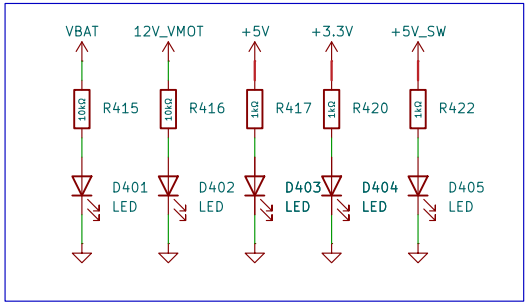

## RGB LED




## Servo outputs




## LED indicators



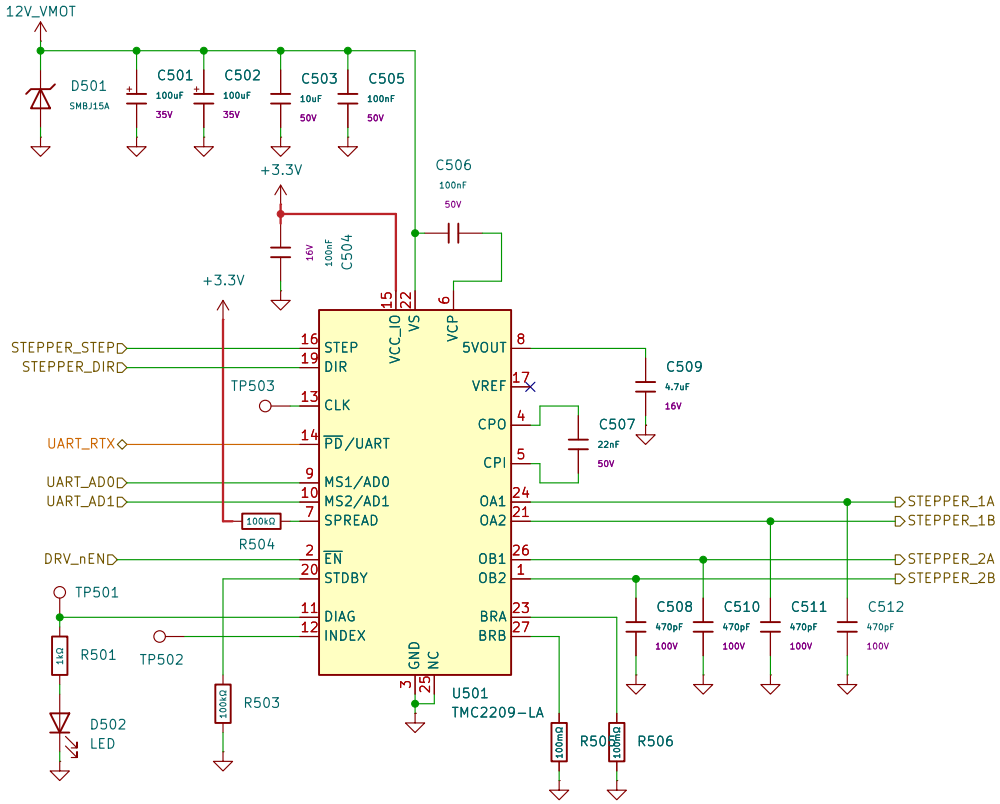

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Sheet: /IO/

File: 0400_IOEXT.kicad_sch

**Title: IO Adaptation Stage**

| Size: A4 | Date: 2026-02-10 | **Rev: B1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 4/8 |




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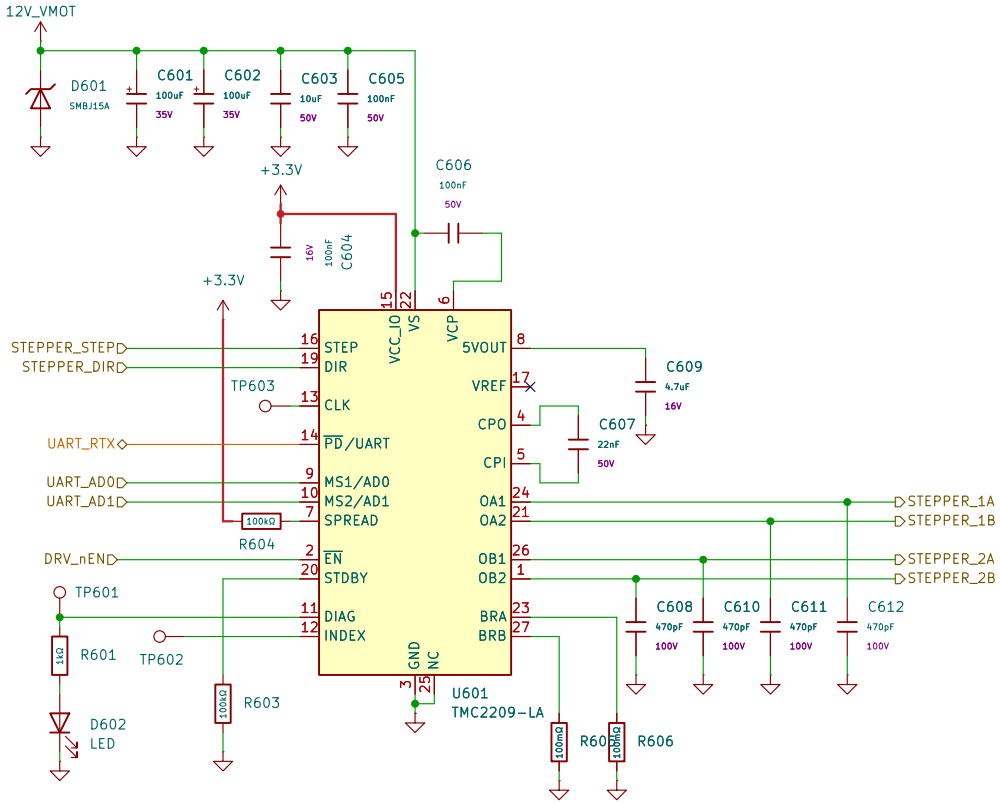
Sheet: /Motor1/

File: 0200_Motors.kicad_sch

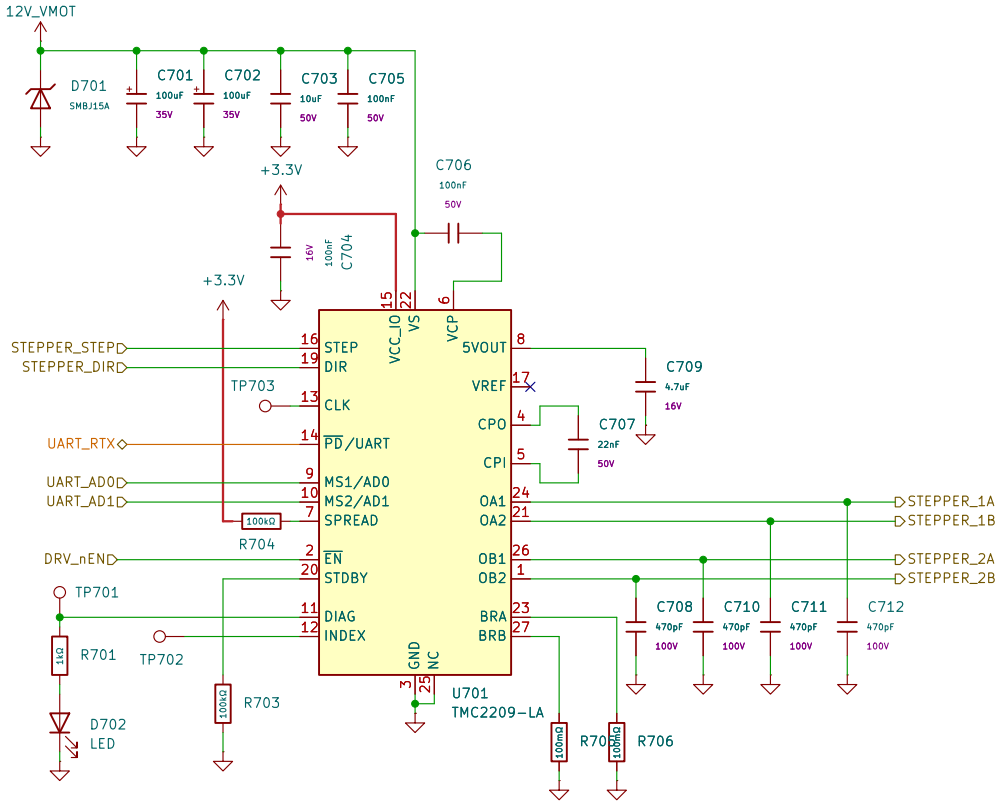
**Title: Stepper Motor**

Size: A4 | Date: 2026-02-10 | **Rev: B1**

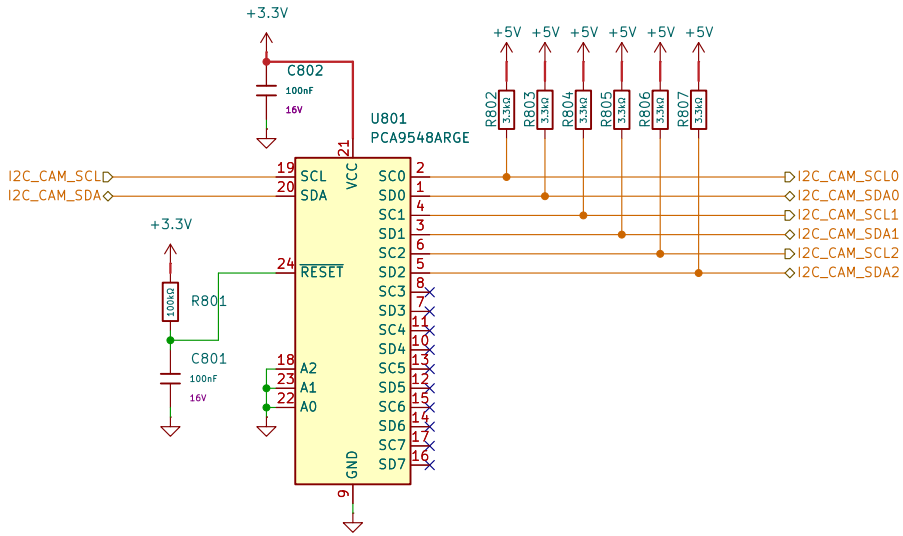
KiCad E.D.A. 10.0.1 | Id: 5/8




| | | |
|---|---|---|
| Les Opossums – Gabriel de Melo Souza | | |
| Sheet: /Motor2/ | | |
| File: 0200_Motors.kicad_sch | | |
| **Title: Stepper Motor** | | |
| Size: A4 | Date: 2026-02-10 | **Rev: B1** |
| KiCad E.D.A. 10.0.1 | | Id: 6/8 |




| | |
|---|---|
| Les Opossums – Gabriel de Melo Souza | |
| Sheet: /Motor3/ | |
| File: 0200_Motors.kicad_sch | |
| **Title: Stepper Motor** | |
| Size: A4 | Date: 2026-02-10 |
| KiCad E.D.A. 10.0.1 | Rev: B1 |
| | Id: 7/8 |




| | | |
|---|---|---|
| Sheet: /I2C_CAM/ | | |
| File: 0300_I2C_CAM.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: 2026-02-10 | **Rev:** |
| KiCad E.D.A. 10.0.1 | | Id: 8/8 |